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Phase 2 Progress Report
QuizMaster

About QuizMaster

It is a command-line quiz application using Go. It allows users to attempt multiple choice questions, choose the difficulty level, get quick feedback, and track their previous results.

Key features

a) Difficulty levels (easy, medium, hard)
b) Multiple choice questions
c) Answer validation (accepts numeric and alphabetic inputs)
d) Instant feedback
e) Score calculation (displays final score, percentage, and performance feedback)
f) Quiz history (saves previous attempts including score, difficulty level, dates)
g) Leaderboard

Technologies used

a) Go (main programming language used)
b) Visual studio code
c) JSON (stores quiz questions and attempt history)
d) Go standard library (handles input/output, file operations etc)
e) Git / GitHub (for project hosting)

Implementation

a) main.go - provides the menu driven interface
b) models.go - defines Question and Attempt structures.
c) questions.json - stores multiple choice questions and their difficulty levels.
d) questions.go - loads questions from JSON and validates data.
e) quiz.go - handles question display, answer validation, feedback, and score calculation.
f) history.go - saves, loads, displays previous quiz attempts using history.json

Current progress

The core modules, including project setup, question management, quiz execution, score calculation, and history management, have been implemented and tested. The remaining work includes leaderboard implementation, additional testing, and final integration.